

Short Read Sequencing Workshop Day 2

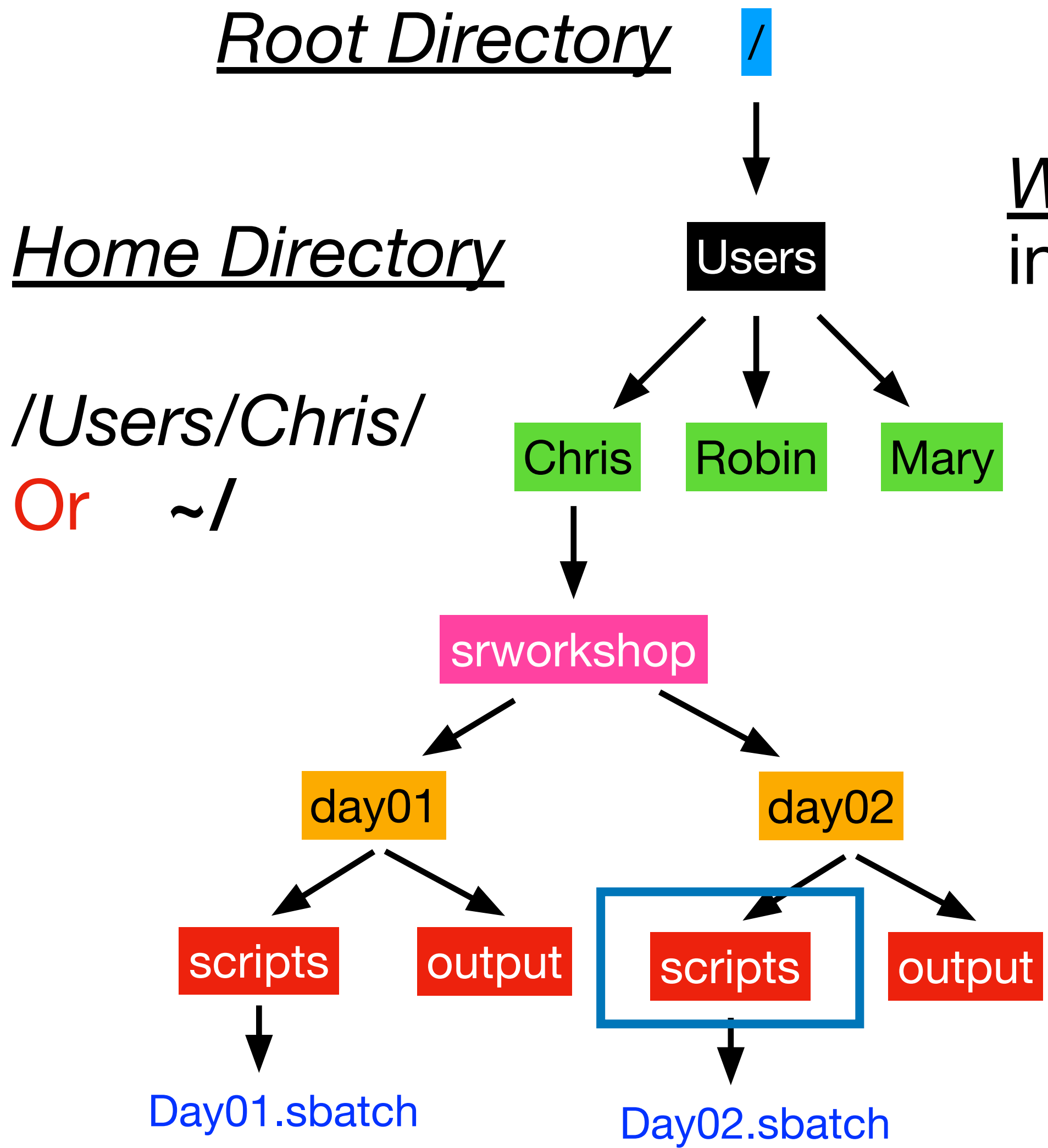
Linux

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Navigating the Hierarchical Directory Structure

Linux Operating System File Structure



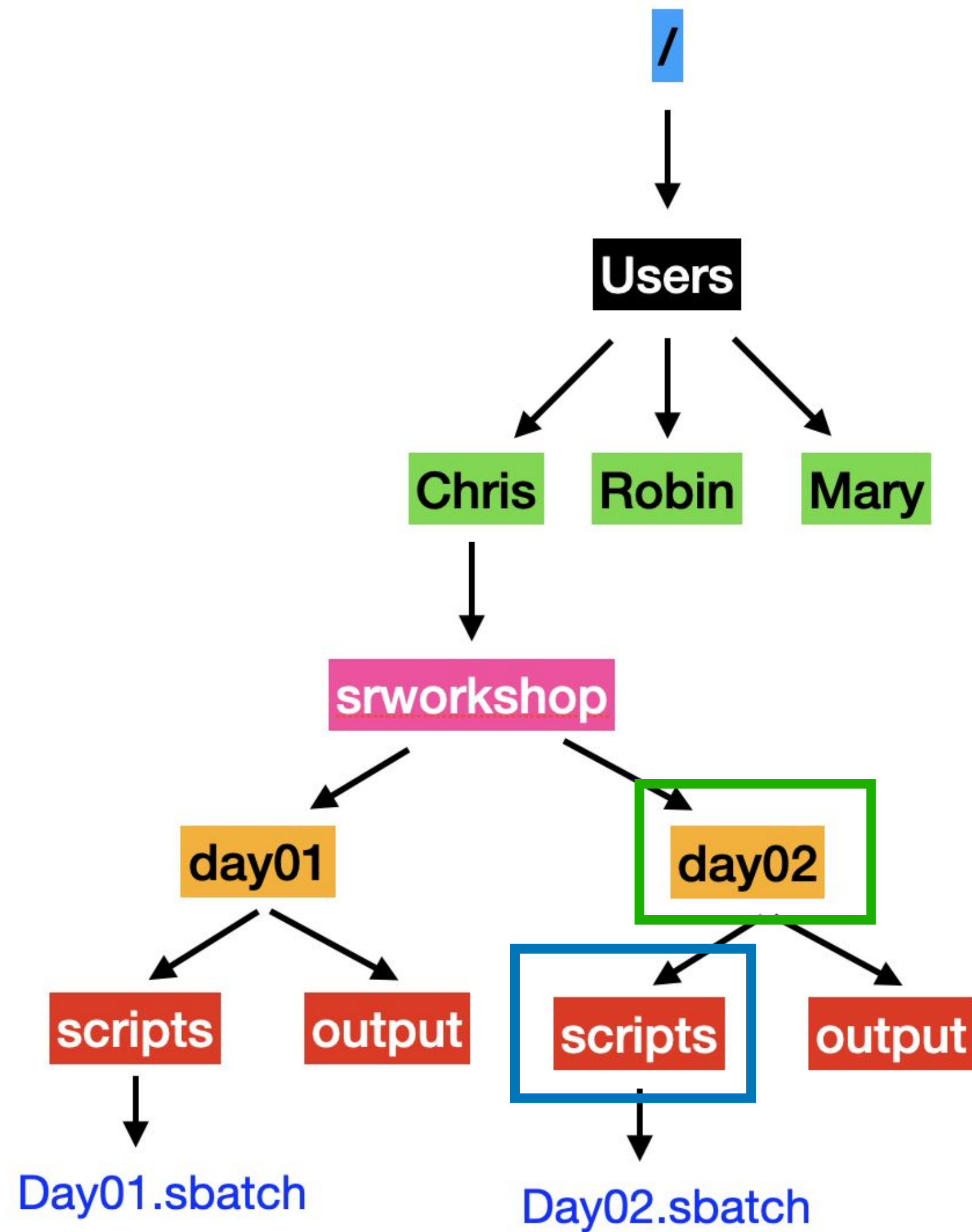
Working Directory: Current Directory you are in

/Users/Chris/srworkshop/day02/scripts

or *./*

Shortcut for current working directory

Navigating the Hierarchical Directory Structure



Objective: Change from current working directory of **scripts** to **day02**

1. Absolute Path - path from root directory

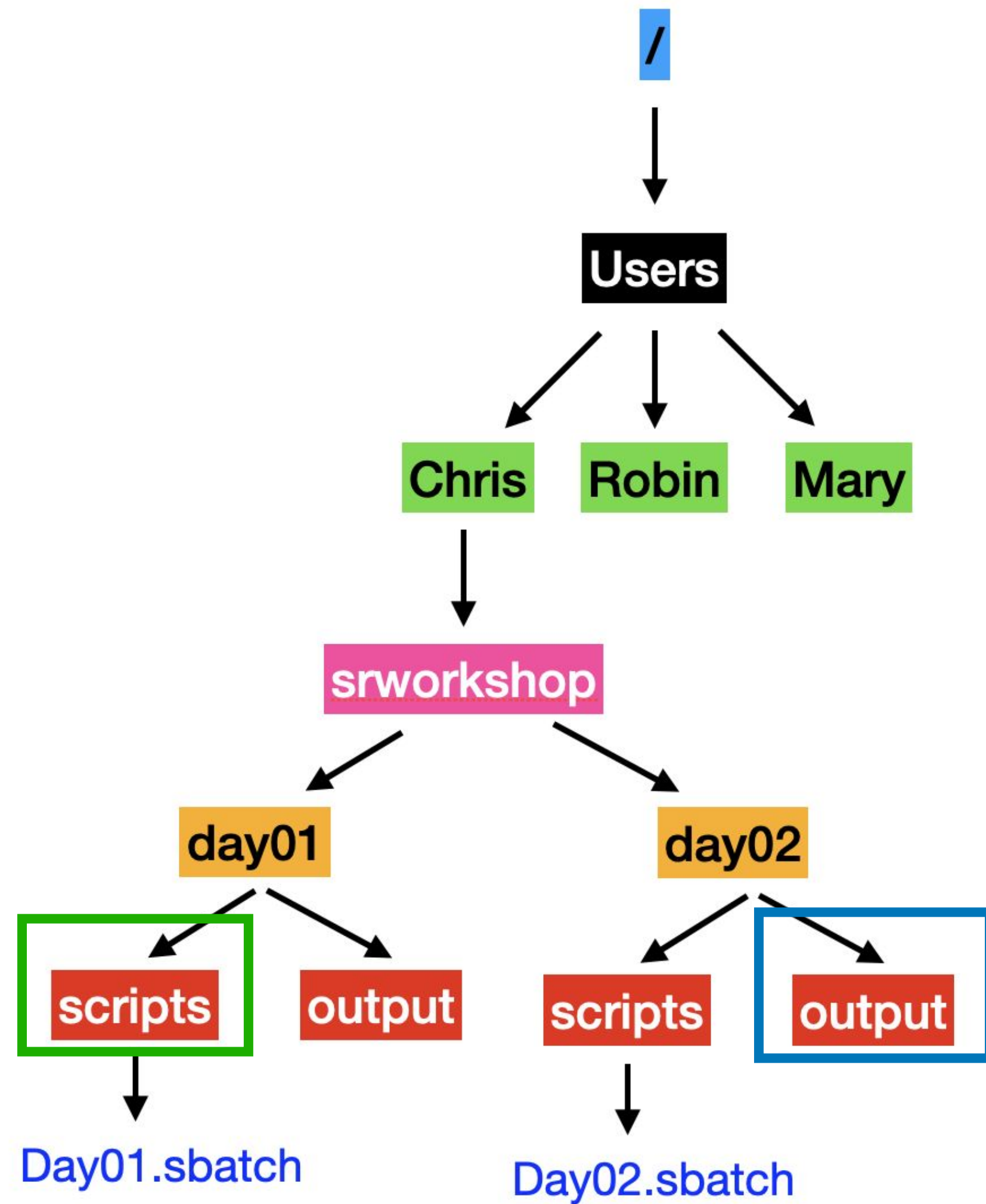
```
cd /Users/Chris/srworkshop/day02
```

2. Relative Path - path from working directory

```
cd ../
```

Move to parent directory (one directory above)

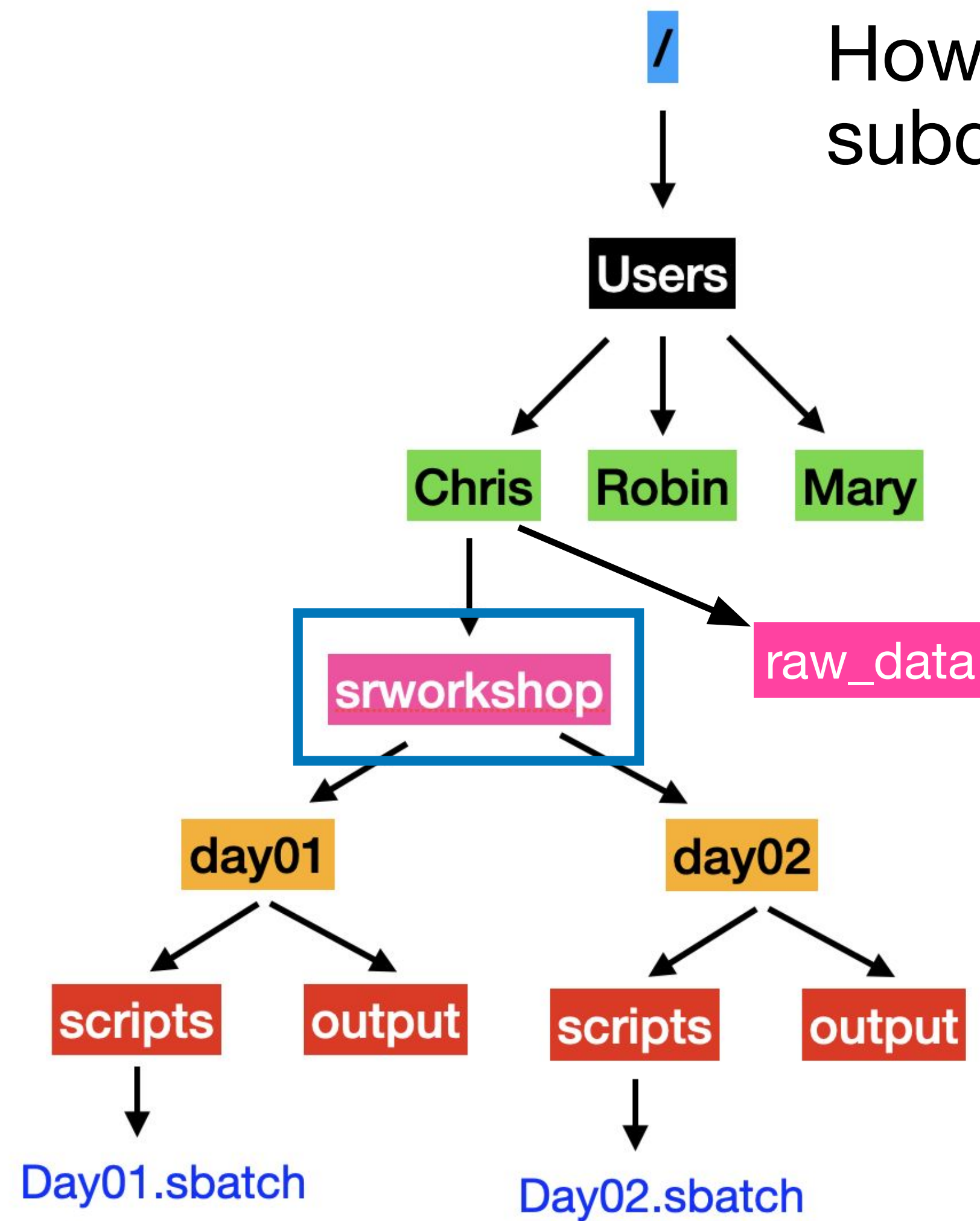
Navigating the Hierarchical Directory Structure



Write a command to navigate from day02 output to day01 scripts using a *relative path*

```
cd ../../day01/scripts
```

Creating the Hierarchical Directory Structure



How can we create a new directory called **raw_data** as a subdirectory of **Chris** from the current working directory:

/Users/Chris/srworkshop

1. Absolute Path

`mkdir /Users/Chris/raw_data`

2. Relative Path

`mkdir ../raw_data`

3. Change Directories to Parent

`cd ..`

`mkdir raw_data`

Unix Command Anatomy

command [options] [arguments]

└ also called [flags]

The executable program you want to run (ls, grep, mkdir). It always is first and is case-sensitive.

Modifiers that change how the command behaves. They are typically prefixed with a single dash followed by a single-letter (like -l).

The target or data the command operates on. These are usually file names, directory paths, or text strings. Flags can also require arguments.

Unix Command Anatomy

command **[options]** **[arguments]**

How do we write a command that shows all files in the long format?

1. `ls -a -l <directory>`

2. `ls -al <directory>` ★

3. `ls --all -l <directory>`

ls Option	Description
a	Lists all files, including hidden files beginning with a .
l	Displays results in the long format, displays permissions
r	Displays results in reverse order (Normally, ascending alphabetical)
h	In the long format, displays the human readable size of the file

Trick question: they are all correct!

Unix Command Anatomy

```
command [options] [arguments]
```

What are the components of this command?

```
ls -al <directory>
```


Unix Command Anatomy

man ls

To Do: Use the `man` command to find a complete list of the `ls` options for the following

1. On the AWS, navigate to `/Users/<username>/srworkshop/day02`
2. Print the contents of the directory to show all files in the long format
3. Now, print the contents of the directory to show the index number of each file or directory



Unix Command Anatomy

1. Combining commands with the pipe operator

```
command1 [input] | command2
```

For any command in the bash command line, you can use the “|” command to have the output of the first command become the input of the second command

2. Saving outputs to files with redirect

```
command1 [input] > output.file
```

The “>” takes the output of a command series and puts it into a new output file

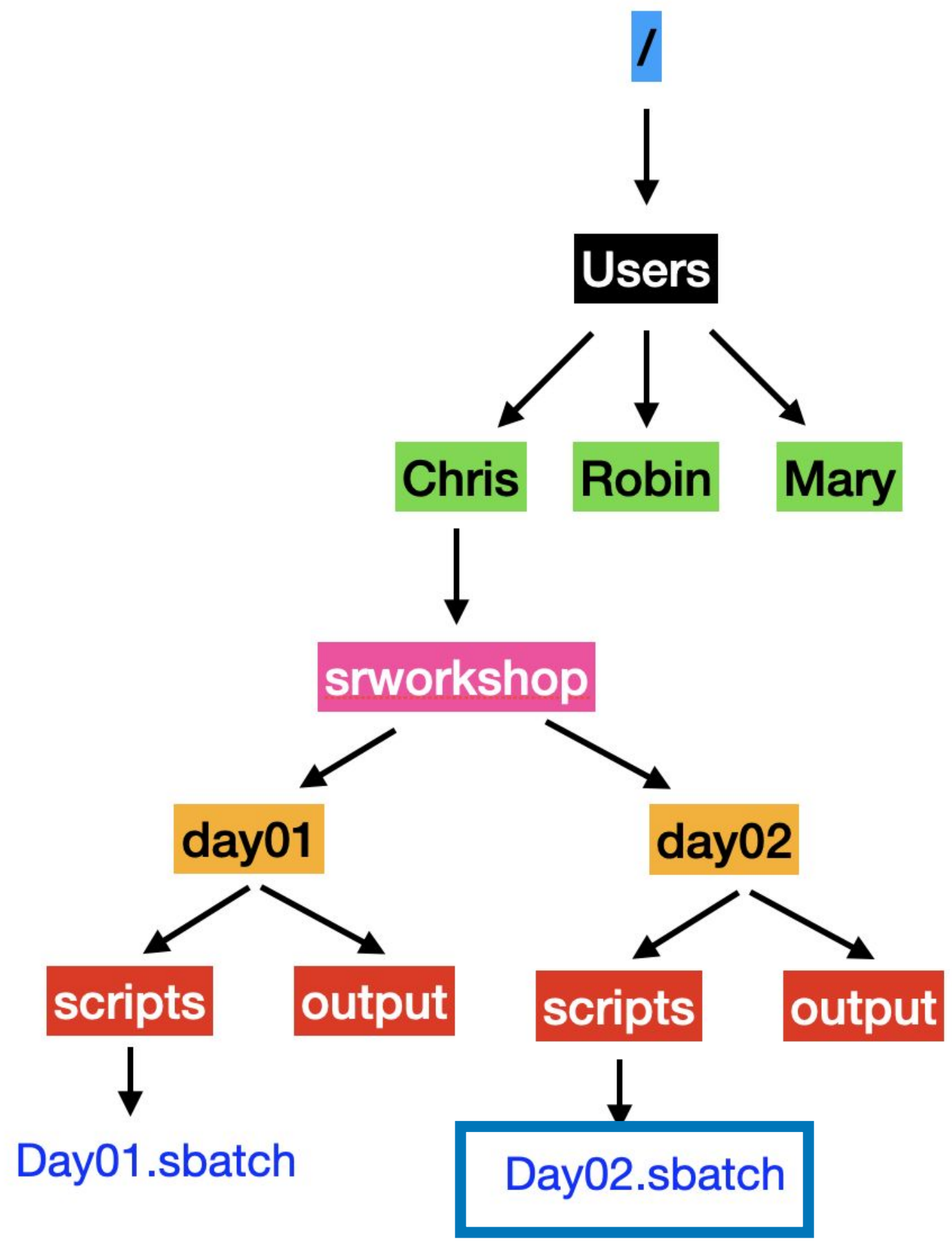
Complete Worksheet day02 worksheet.md Part 1



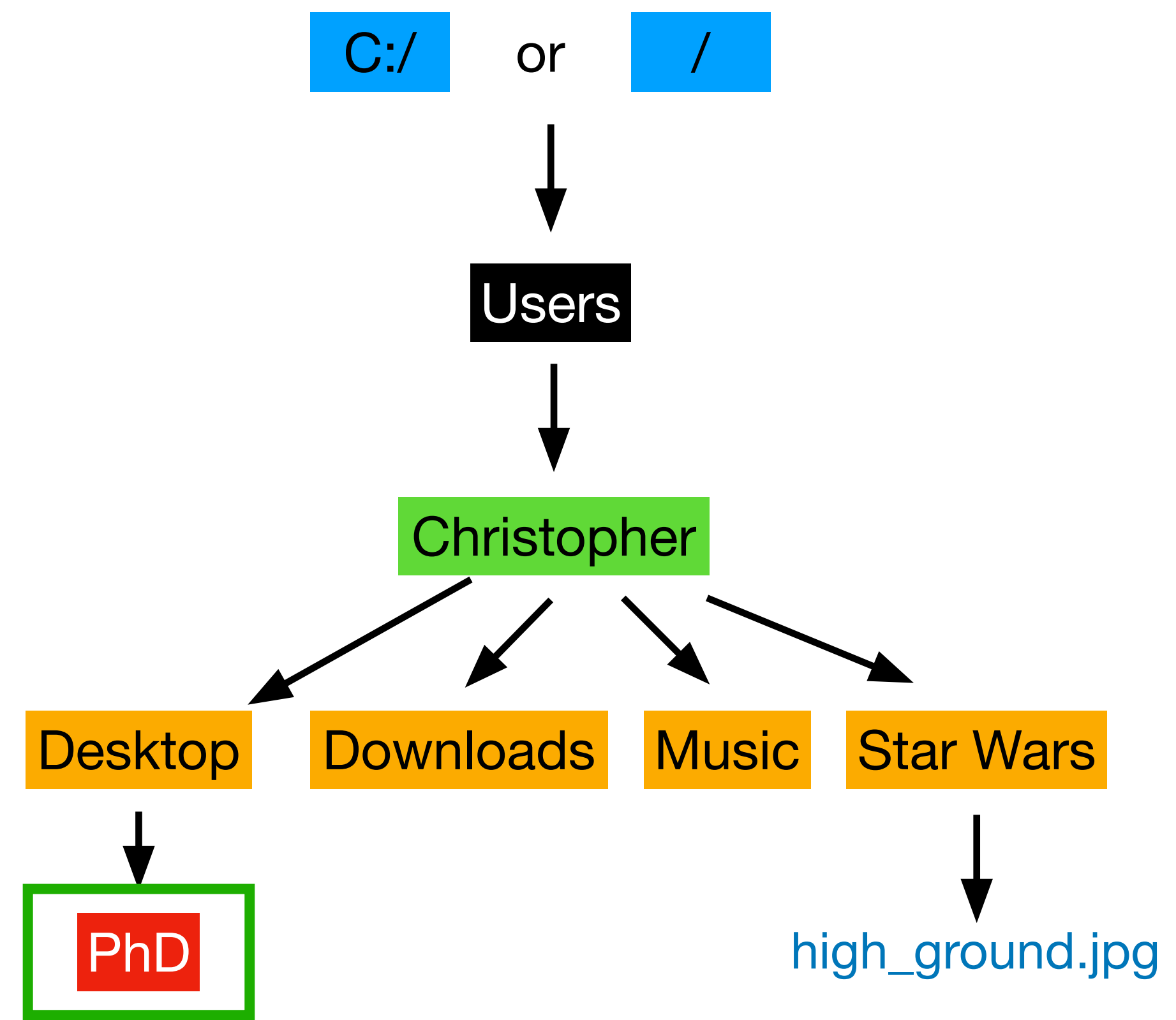
Note: We will be using the **/scratch/** directory rather than home directory

Copying a File from the Super Computer to the Local

AWS Super Computer



Local Computer



How do I copy Day02.sbatch and move it to my local PhD Directory?

Copying a File from the Supercomputer to the Local

```
rsync <source> <destination>
```

On local computer go to target destination directory

```
local $ cd Desktop/PhD
```

```
local $ rsync <username>@13.57.133.102:/Users/<username>/srworkshop/day02/scripts/Day02.sbatch ./
```

AWS login info
(username@ip)

:<absolute_path_source>

Relative path
destination local

Permissions

A file or directory's permissions can be viewed using `ls -l` and it can be altered using `chmod`

`chmod <permission code> <file>`

```
nbratset@srAWS:~/srworkshop$ ls -l
total 88
-rw-r--r-- 1 nbratset nbratset 35149 Jul  6 14:48 LICENSE
-rw-r--r-- 1 nbratset nbratset  1364 Jul  6 14:48 README.md
drwxr-xr-x 6 nbratset nbratset  6144 Jul  6 14:48 day01
drwxr-xr-x 6 nbratset nbratset  6144 Jul  6 14:48 day02
drwxr-xr-x 6 nbratset nbratset  6144 Jul  6 14:48 day03
drwxr-xr-x 6 nbratset nbratset  6144 Jul  6 14:48 day04
```

drwxrwxrwx

d = Directory

r = Read

w = Write

x = Execute

chmod 777

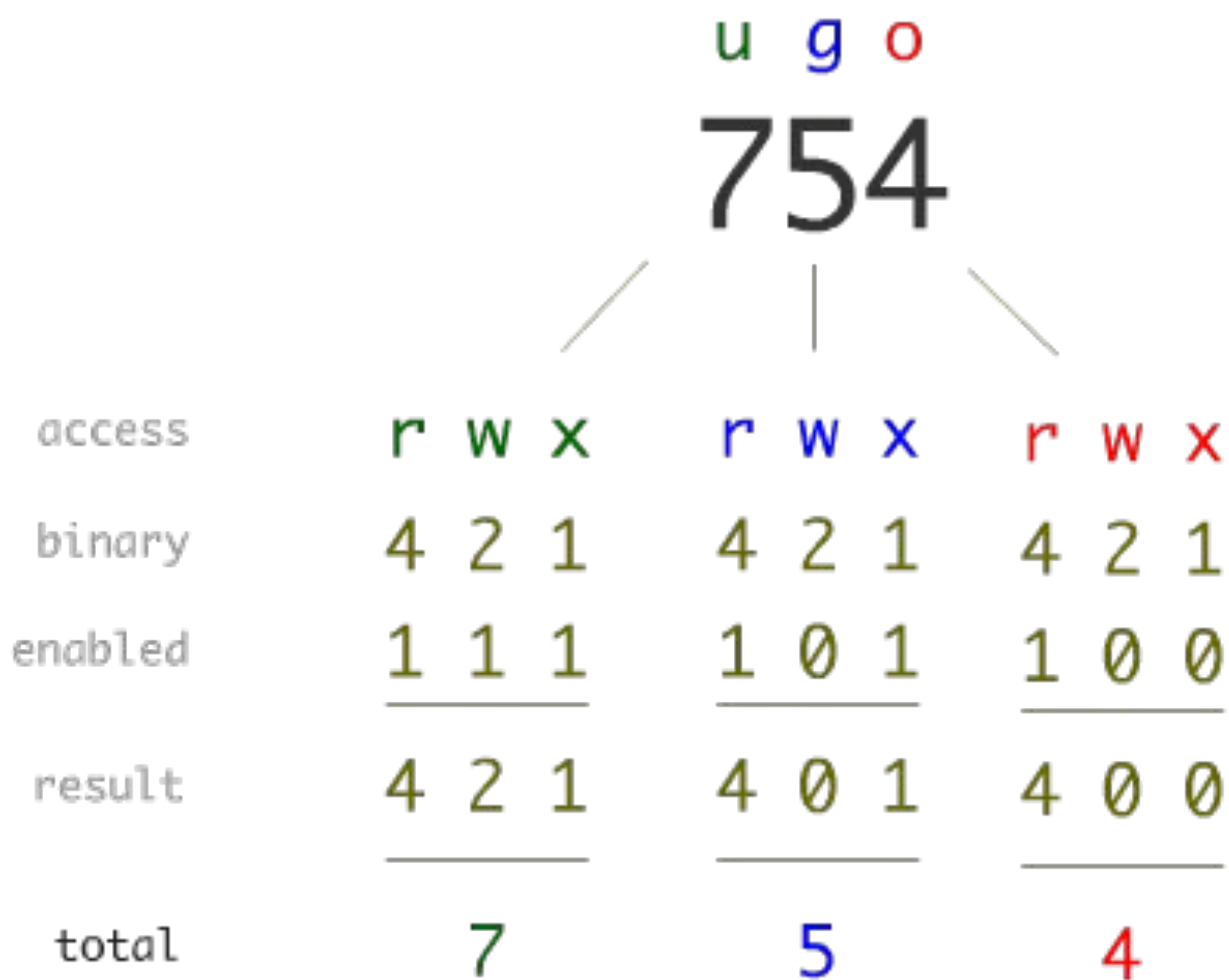
rwX | rwX | rwX
Owner | Group | Others

7	rwX	111
6	rw-	110
5	r-X	101
4	r--	100
3	-wX	011
2	-w-	010
1	--X	001
0	---	000

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`drwxrwxrwx`

d = Directory
r = Read
w = Write
x = Execute

`chmod 777`

`rwX | rwX | rwX`
Owner | Group | Others

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6	rw-	110
5	r-X	101
4	r--	100
3	-wX	011
2	-w-	010
1	--X	001
0	---	000

Complete Worksheet day02 worksheet.md Part 2

